

Substance Abuse and Mental Health Among Racial Groups in Canada, with a Focus on Black Canadians

1. Introduction

This report explores patterns in mental health and substance use indicators across racial groups in Canada using data from 2021 to 2023. The analysis focuses on perceived mental health status and a broad set of clinical indicators, including mood disorders, anxiety, substance use disorders, and service utilization. By conducting chi-square tests of independence and examining standardized residuals, we identify specific mental health conditions where certain racial groups—particularly Black individuals—show disproportionate representation. This report aims to highlight disparities in mental health prevalence and service access, with a focus on identifying potential structural and social inequities affecting racialized populations.

2. Data & Methods

2.1 Data Sources

- **Table 45-10-0080-01 (Perceived Mental Health)**
Quarterly data from Q1 2021 to Q4 2023 for Canadians aged 15+, capturing self-perceived mental health (“Excellent,” “Good,” “Fair/Poor”) across gender, age, and other demographics.
- **Table 13-10-0930-01 (Mental Health Indicators by Population Group)**
Derived from the 2022 Mental Health and Access to Care Survey (MHACS) and released in December 2024. This dataset includes lifetime and 12-month prevalence of disorders such as depression, anxiety, substance use, suicidal ideation, psychosis, stress, and mental health service utilization, stratified by population group (e.g., Black, South Asian, etc.).

2.2 Data Preparation

- Datasets were grouped on **Race**, **Year**, **Quarter/Age Group**, and **Gender** to aggregate values and perform statistical tests .
- Some values regarding proportions marked as dubious or to be used with caution were not included in any statistical testings, but were used for visualization purposes to identify trends.

2.3 Indicator Grouping

Indicators were categorized into:

1. **Mood & Anxiety Disorders** (e.g., depression, bipolar, PTSD)
2. **Substance Use Disorders** (alcohol, cannabis, other drugs)
3. **Suicidality & Stress** (suicidal thoughts, self-rated work stress)
4. **Service Utilization** (professional mental health help)
5. **Summary Indicators** (overall prevalence of any disorder)

2.4 Statistical Analyses

1. Chi-Square Tests of Independence

Constructed **race × indicator** contingency tables using weighted counts. Chi-square tests ($\alpha=0.05$) assessed whether mental health condition distributions varied by race.

2. Standardized Residuals

For significant chi-square tests, residuals were calculated to pinpoint race/indicator cells with unexpected prevalence. Results were scaled by a factor of 100 for interpretability. A residual > 2 indicated statistically significant overrepresentation.

3. Comparative Analyses for Black Respondents

Residuals for Black individuals were examined to determine:

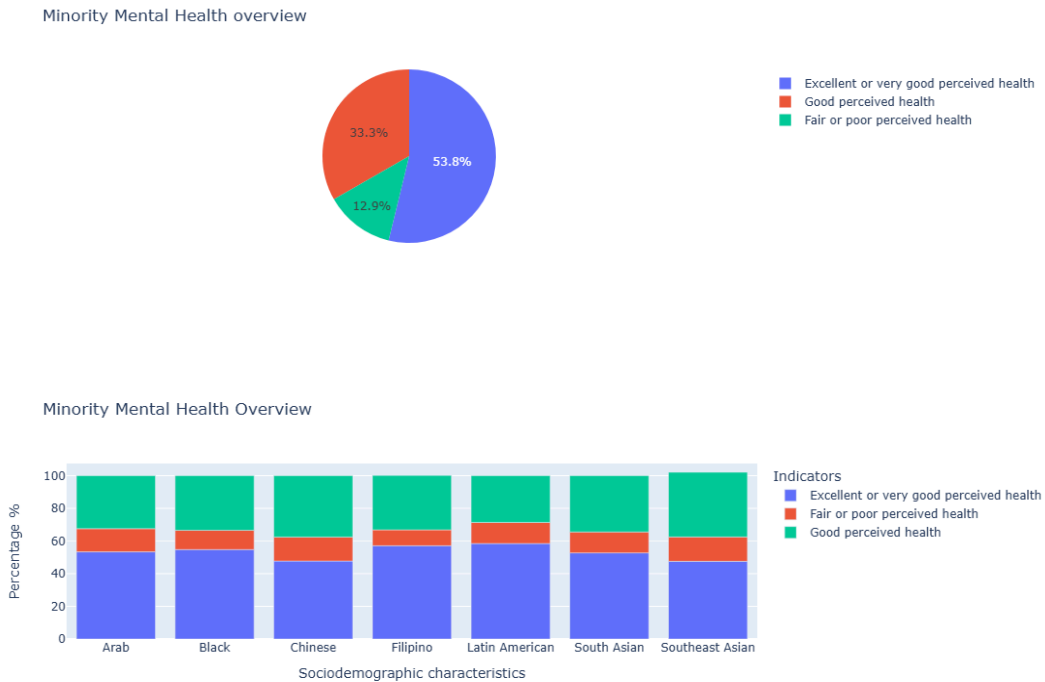
- Indicators with the highest residual values (most disproportionate representation)
- Statistically significant residuals (>2)

4. Visualization & Reporting

Common charts and graphs were generated to illustrate trends in self-reported and mental health indicators across races and time. Heat maps were generated to study residual discrepancies between racial groups, but not all were included to ensure brevity.

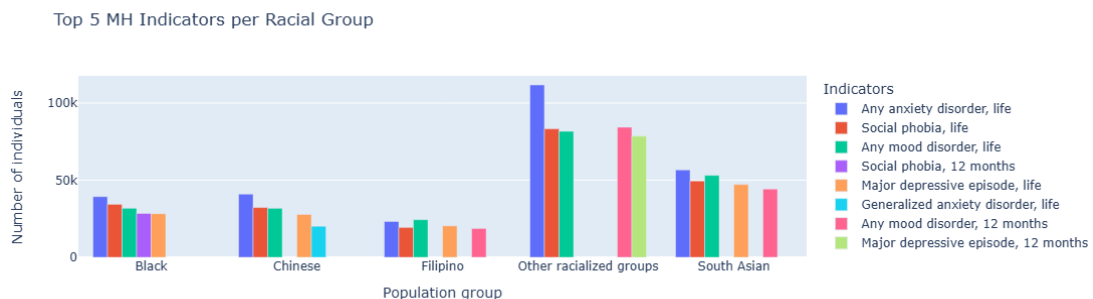
3. Results

3.1 Perceived Mental Health



- Black respondents reported perceived mental health aligned closely with national trends.
- However, standardized residuals reveal higher-than-expected “Poor/Fair” mental health ratings for Black respondents in several quarters, indicating potential disparities in self-perceived wellbeing. These are explored further below.

3.2 Mood & Anxiety Disorders



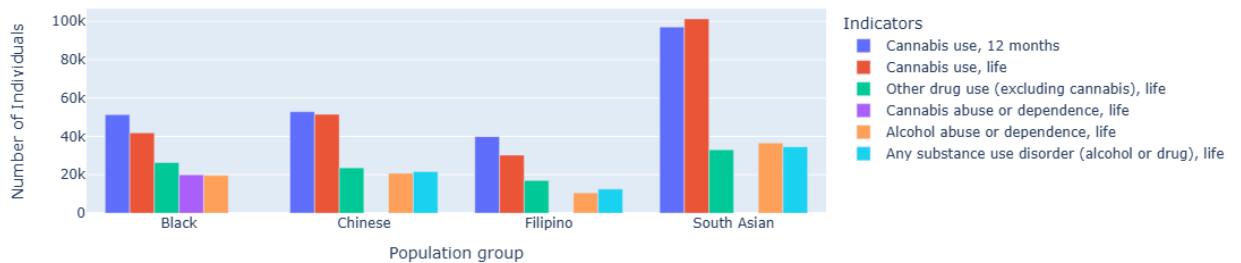
Top 5 mental health indicators for Black Canadians



- Chi-square tests confirm significant racial differences in prevalence.
- Black respondents showed significantly elevated residuals (>2) for 12-month anxiety disorders and lifetime anxiety disorder—indicating overrepresentation beyond what independence would predict.

3.3 Substance Use Disorders

Substance Use across Racial groups

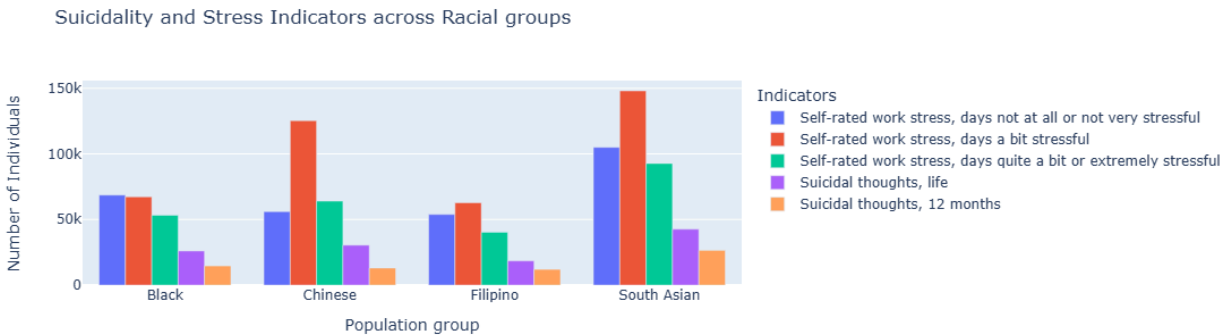


Top 5 Substance Used for Black Canadians



- Prevalence of alcohol and cannabis use disorder varied significantly by race.,
- Black respondents primarily indulged in cannabis use, however, respondents (15-44 y/o) displayed higher-than-expected residuals for other non-cannabis drug use, suggesting a notable disparity in substance-related experiences.

3.4 Suicidality & Stress

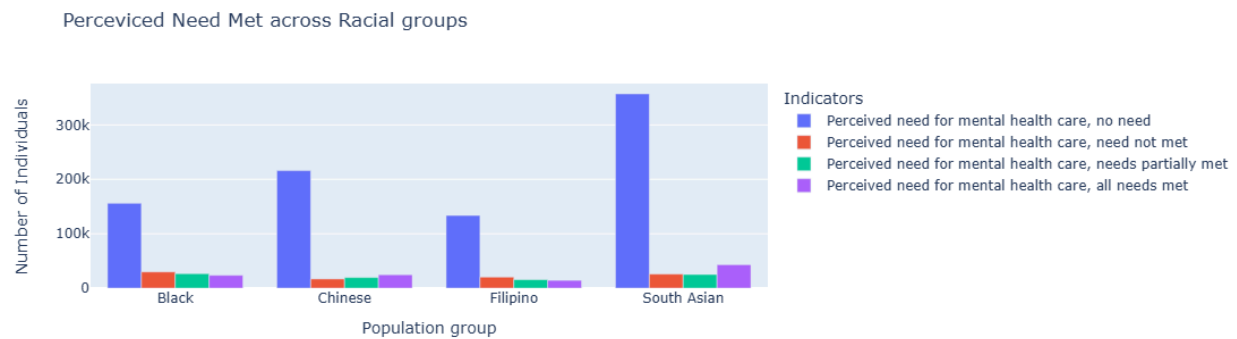


Top 5 Stress and Suicidal Indicators Used for Black Canadians

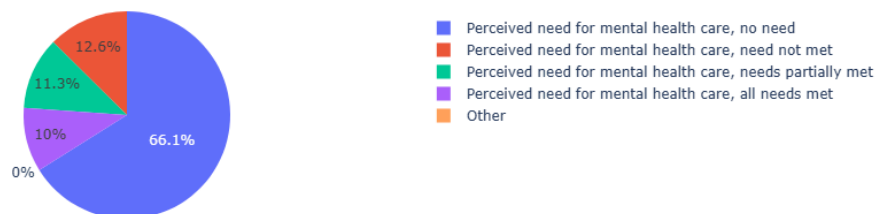


- Racial differences in suicidal ideation and stress were significant.
- However, residuals for Black respondents did not exceed statistically significant thresholds, suggesting alignment with population expectations in these areas.

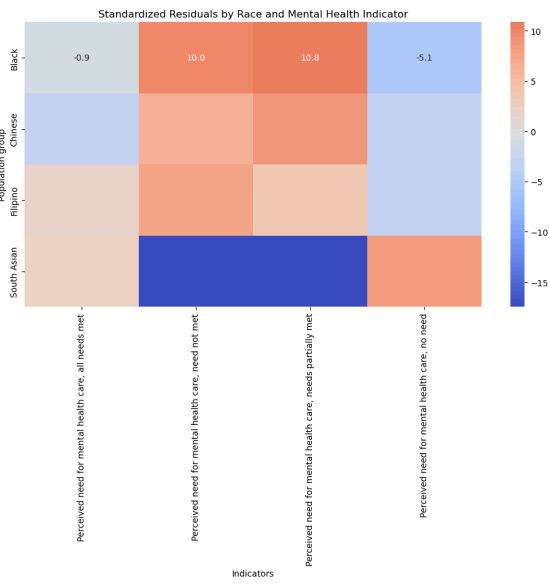
3.5 Perceived Needs Met



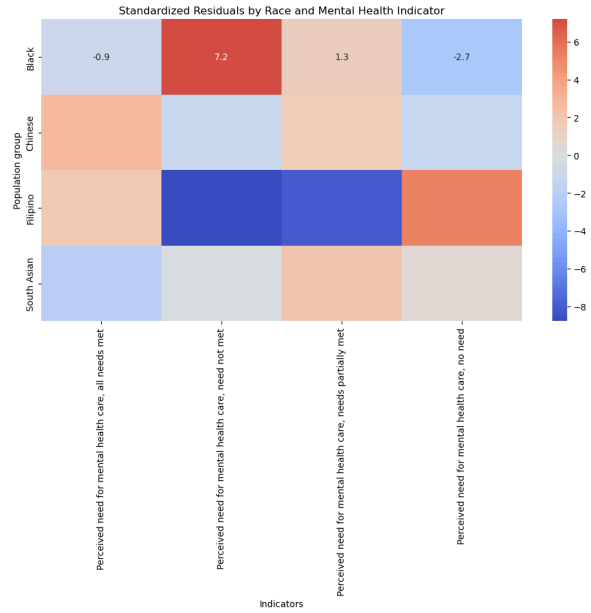
Perceived Need Met for Help for Black Canadians



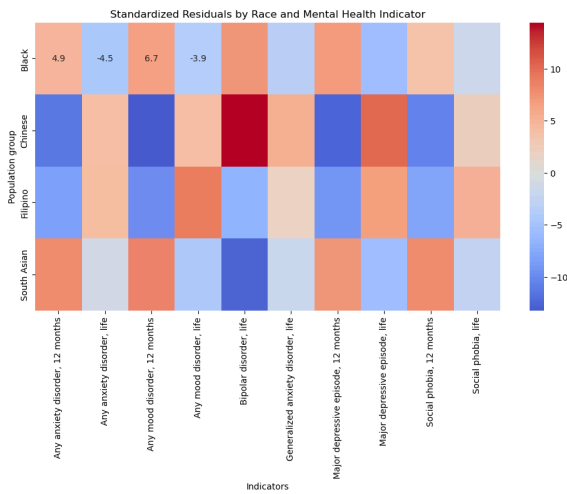
- Racial differences in perceived needs unmet or partially met were significant.
- Residuals for Black respondents with **unmet or partially met** mental health needs exceeded the threshold for statistical significance (> 3), particularly among youth (15–24) and adults (25–44). The disparity was most pronounced for Black Canadians.



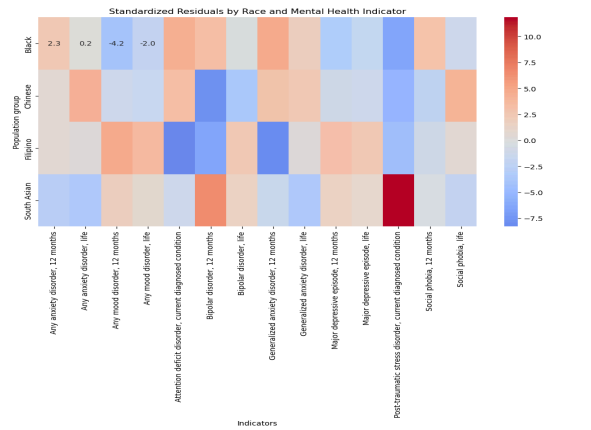
Perceived Need Met Adult (25-44)



Perceived Need Met Youth (15-24)



Mental Health Indicators for Adults (25-44)



Mental Health Indicators for Youth (15-24)

4. Discussion

- **Disparities Evident in Clinical Indicators:** Overrepresentation of anxiety disorders, PTSD, and substance use among Black respondents signals potential structural vulnerabilities (e.g., discrimination, socioeconomic exclusion, access barriers).
 - **Self-Perception vs. Clinical Outcomes:** While “Poor” perceived mental health ratings were elevated, service utilization was not disproportionately different—suggesting unmet needs or access barriers.
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5. Conclusion & Recommendations

- **Black youths** in Canada exhibit disproportionate rates of anxiety disorders, PTSD, and substance use compared to expectations—highlighting areas for targeted public health intervention.
- Young adults exhibited elevated **work-related stress** and **perceived unmet mental health** needs relative to expectations, suggesting potential systemic bias or clinical prejudice.
- Further research should explore **intersectional factors** (income, gender, immigration status) and implement **qualitative studies** on service access and stigma.
- Enhanced surveillance and culturally responsive service planning are critical to addressing these disparities.

Importantly, there is a pressing need for **more racially disaggregated and geographically detailed data**, especially for **newcomer and immigrant populations**, to better capture the diversity of mental health experiences across Canada. This would enable more **nuanced and equitable policy-making**, tailored to the specific needs of underrepresented communities.